Supporting machine-learning of bioactivity data

Noel O'Boyle

Chemical Biology Resources Team Leader

UCL-SGC ML Networking Event
17/10/2025

EMBL-EBI

The European Molecular Biology Laboratory

EMBL-EBI

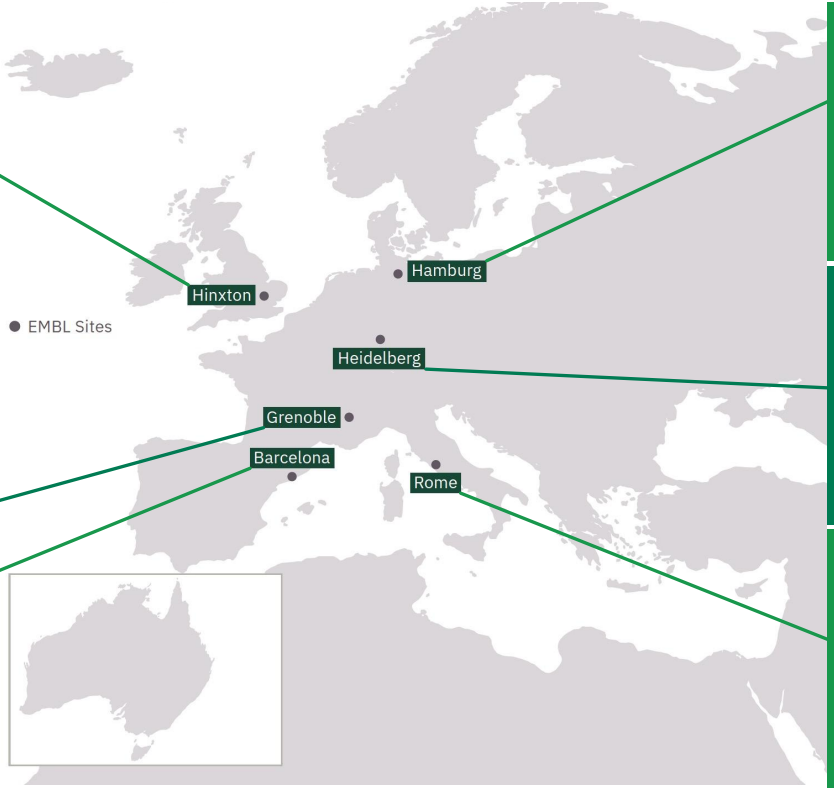
Bioinformatics

Grenoble

Structural biology

Barcelona

Tissue biology and disease modelling

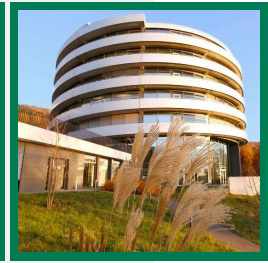


Hamburg

Structural biology

Heidelberg

Life sciences

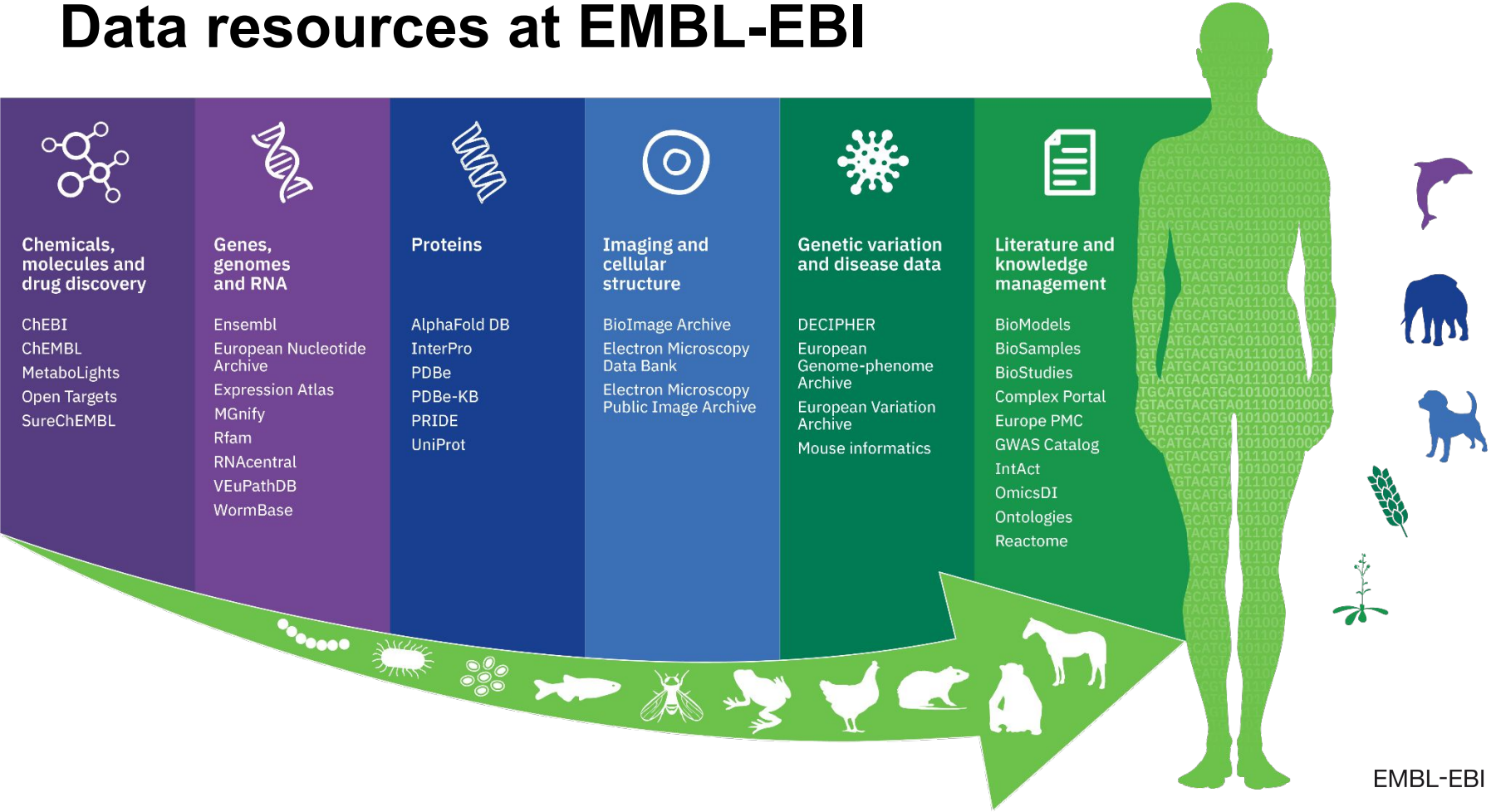


Rome

Epigenetics and neurobiology

EMBL-EBI

Data resources at EMBL-EBI



EMBL-EBI

Chemical Biology Services @ EMBL-EBI

ChEMBL

Bioactivities from literature and deposited

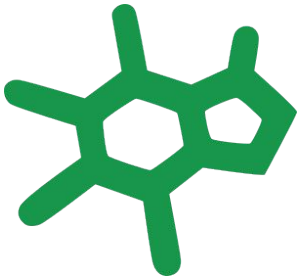
Drug/candidate information

SureChEMBL

Structures from patents

UniChem

Link databases via InChI



ChEBI

Chemical ontology

Chemical registration db for biological dbs

OPSIN

Convert IUPAC names to structure

ChEMBL 36 (Sept 2025)

17,803
Targets

2,878,135
Distinct Compounds

24,267,312
Activities

99,142
Publications

336
Deposited Datasets

New data source: Chemical probe data from the scientific literature (manuscript in prep.)

Updated data sources: (selected)

- Medicinal chemistry literature
- BindingDB bioactivities from patents (was 4K assays, now 14K)
- EMBL Heidelberg Gut Microbiome Drug Interactions
- AI-driven Structure-enabled Antiviral Platform (ASAP)
- SGC EUbOPEN chemogenomic library

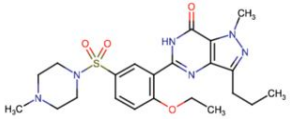
ChEMBL Drug and Clinical Candidate data

Drug Development

- ClinicalTrials.gov
- USAN applications
- INN applications
- ATC classification

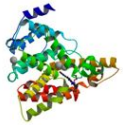
Drug Approval

- BNF
- FDA Orange Book
  FDA New Drugs
- NEW: EMA Drugs
- DailyMed
- Withdrawn Drugs



Name, synonym & chemical structure mapping

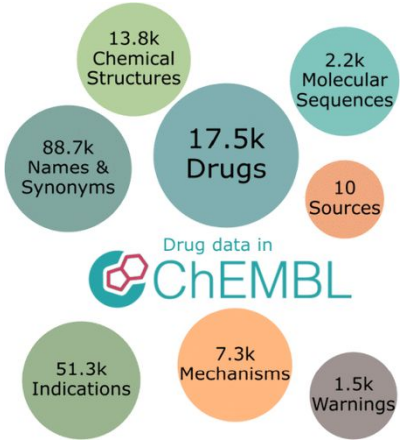
Drug property annotation
NEW: orphan flag



Mechanism annotation

Disease annotation

Warning annotation



Development of a drug side-effects resource

Just funded!

Open Targets

EMBL-EBI

Fiona Hunter et al. **Drug and clinical candidate drug data in ChEMBL.** *J. Med. Chem.* **2025**, *68*, 19800.

Recent articles that highlight ChEMBL data usage

Journal of Chemical Information and Modeling > Vol 64/Issue 24 > Article

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CHEMICAL INFORMATION | December 10, 2024

Influence of Data Curation and Confidence Levels on Compound Predictions Using Machine Learning Models

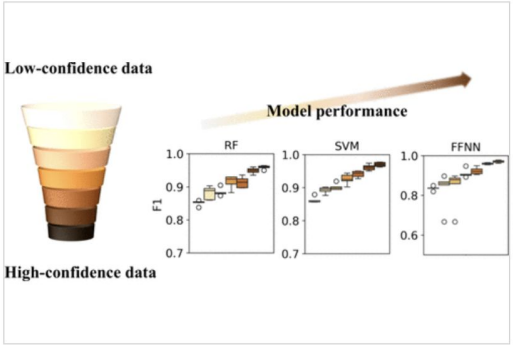
Elena Xerxa, Martin Vogt, and Jürgen Bajorath*

Open PDF

Supporting Information (1)

Abstract

While data curation principles and practices are a major topic in data science, they are often not explicitly considered in machine learning (ML) applications in chemistry. We have been interested in evaluating the potential effects of data curation on the performance of molecular ML models. Therefore, a sequential curation scheme was developed for compounds and activity data, and different ML classification models were generated at increasing data confidence levels and evaluated. Sequential data curation was found to systematically increase classification performance in an incremental manner due to cumulative effects of individual data curation criteria. The analysis of chemical space distributions of compound subsets at different data confidence levels revealed that the separation of compounds with different class labels in chemical space generally increased during



- Models improve with carefully selected (“curated”) input

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Recent articles that highlight ChEMBL data usage

- Protein variants have a huge impact on bioactivity profiles

Excuse me, there is a mutant in my bioactivity soup!

A comprehensive analysis of the genetic variability landscape of bioactivity databases and its effect on activity modelling

Marina Gorostiola González[1,2,‡], Olivier J. M. Béquignon[1,‡], Emma J. Manners[3], Anna Gaulton[3], Prudence Mutowo[3], Elisabeth Dawson[3], Barbara Zdrazil[3], Andrew R. Leach[3], Adriaan P. IJzerman[1], Laura H. Heitman[1,2], and Gerard J. P. van Westen*[,1]

[1]*Leiden Academic Centre of Drug Research, Leiden University, 55 Einsteinweg, 2333 CC. Leiden, The Netherlands*
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[‡]*These authors contributed equally*
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Journal of Chemical Information and Modeling > Vol 64/Issue 5 > Article
Open Access
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CHEMICAL INFORMATION | February 23, 2024

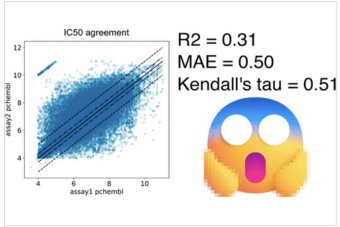
Combining IC_{50} or K_i Values from Different Sources Is a Source of Significant Noise

Gregory A. Landrum*, and Sereina Riniker*

Open PDF | Supporting Information (1)

Abstract

As part of the ongoing quest to find or construct large data sets for use in validating new machine learning (ML) approaches for bioactivity prediction, it has become distressingly common for researchers to combine literature IC_{50} data generated using different assays into a single data set. It is well-known that there are many situations where this is a scientifically risky thing to do, even when the assays are against exactly the same target, but the risks of assays being incompatible are even higher when pulling data from large collections of literature data like ChEMBL. Here, we estimate the amount of noise present in combined data sets using cases where measurements for the same compound are reported in multiple assays against the same target. This approach shows that IC_{50} assays selected using minimal curation settings have poor agreement with each other: almost 65% of



- Combining disparate data may be problematic
- Curation/clean-up of extracted data has a significant positive impact
- Considers bioactivity against variant vs. wild-type targets

EMBL-EBI

How can we help with preparation of datasets?

Provide guidance

- Careful design of queries
- Issues to consider

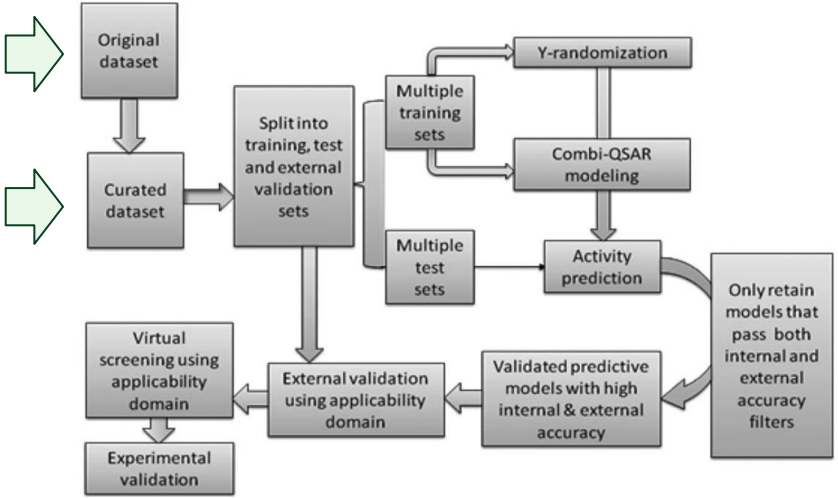
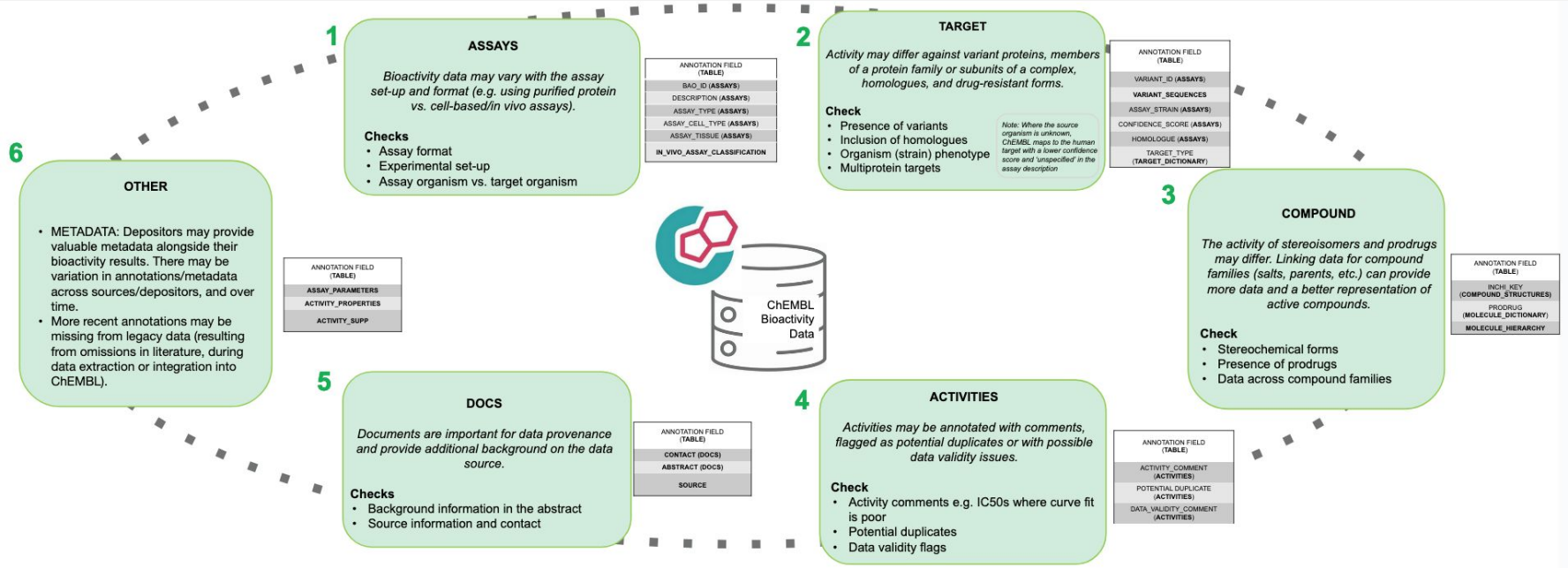


Figure 2. Predictive QSAR modeling workflow.

Best Practices for QSAR Model Development, Validation, and Exploitation – Tropsha. *Molecular Informatics.* **2010**, *29*, 476

What guidelines can we offer for data checks and clean-up?



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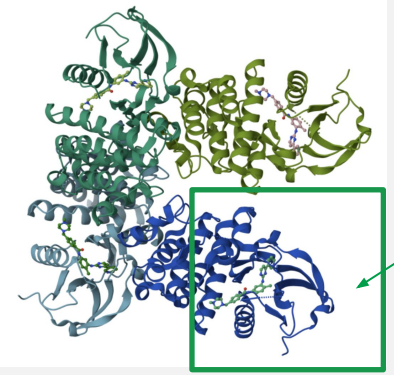
Be careful of protein variants

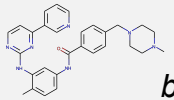
✅ Is my protein target a variant?

- Protein variants can significantly impact compound bioactivity.
- Pooling bioactivity data for wild-type and variant targets can skew averages.

human ABL1	WT + variants	Wild-type only	T315I variant
pIC50 Imatinib	6.57 (n=39)	6.71 (n=30)	5.13 (n=3)

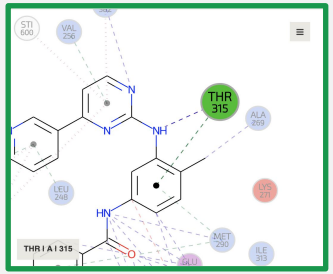


ABL kinase + Imatibib (PDBe; 2hyy)



Imatinib

Imatinib is an approved drug that binds ABL1 kinase



Imatinib binding site

The ABL1 kinase T315 residue forms H-bonds with imatinib, this binding interaction is disrupted in the T315I drug-resistant variant

FIELDS: **ASSAYS**/VARIANT_ID, **VARIANT_SEQUENCES**/ALL

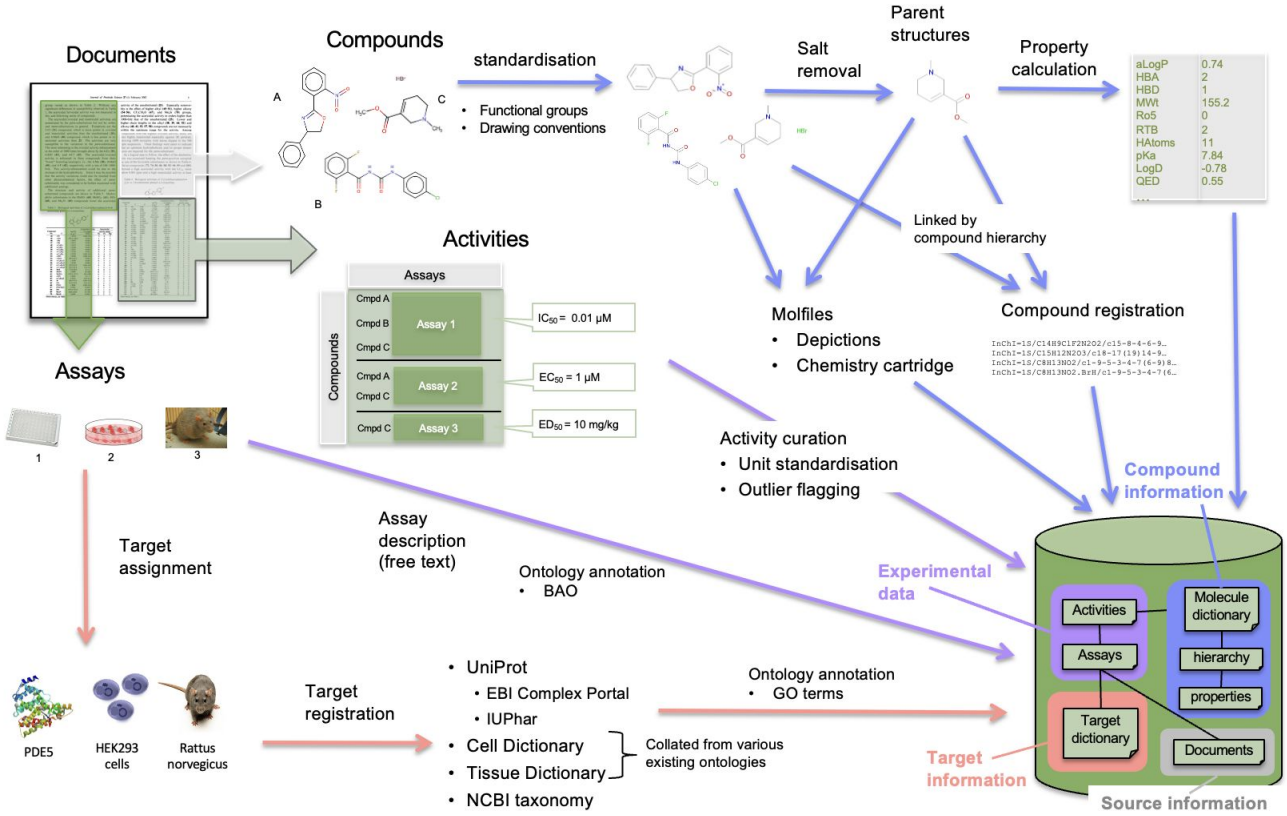
EMBL-EBI

What do we promise our users?

- ✅ Extracted medicinal chemistry literature data disseminated in a structured form
- ✅ Curation and standardisation where appropriate
- ✅ Periodic enhancements and updates (when appropriate, and resource allows)
- ✅ Training and regular publications describing data content, structure, and curation
- ✅ Correction of reported errors in extraction

- ❌ We **do not** peer-review extracted literature and cannot guarantee error-free sources
- ❌ We **cannot avoid** some errors during data ingestion and integration
- ❌ We have **very limited control** over data reuse by our users

Data quality & data curation in ChEMBL



Bento, A.P. et al. **An open source chemical structure curation pipeline using RDKit.** *J. Cheminform.* **2020**, *12*, 51.

https://github.com/chembl/ChEMBL_Structure_Pipeline

Papadatos G. et al. **Activity, assay and target data curation and quality in the ChEMBL database.** *J. Comput. Aided Mol. Des.* **2015,** *29*, 885.

ChEMBL molecule standardiser (RDKit)

Checker
(Run on DB loading)
- empty CTAB
- Polymer
- V3000 mol file
- 3D coord in mol file
- Illegal bond type
- Overlapping atoms
- Coordinates all zero
- Crossed bonds in ring

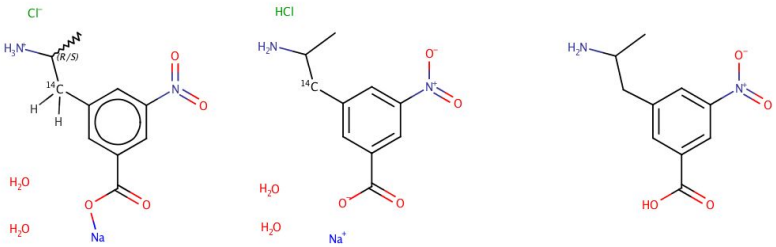
…….

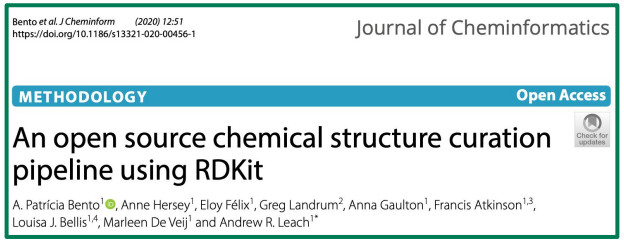
Standardiser
- Clear S Block data
- Kekulise structures
- Fixes bad valence
- Fix KO to K+ O- & NaO to Na+ O-
- NO2 groups to [N+](=O)[O-]
- NH+ Cl- to N and HCl
- Fix wiggly bonds on sp3 C
- Remove explicit Hs
- Straighten triple bonds & allenes
- Standardise sulphoxides

…….

Salt Stripper (GetParent function)
- Salts removed according to CHEMBL salt list
- Solvents removed according to solvents list
- Remove isotopes
- Both components are salts
- Re-neutralise molecule if necessary

……




Bento et al. J Cheminform (2020) 12:51
https://doi.org/10.1186/s13321-020-00456-1

Journal of Cheminformatics

METHODOLOGY Open Access

An open source chemical structure curation pipeline using RDKit

A. Patrícia Bento[1], Anne Hersey[1], Eloy Félix[1], Greg Landrum[2], Anna Gaulton[1], Francis Atkinson[1,3], Louisa J. Bellis[1,4], Marleen De Veij[1] and Andrew R. Leach[1*]

https://github.com/chembl/ChEMBL_Structure_Pipeline

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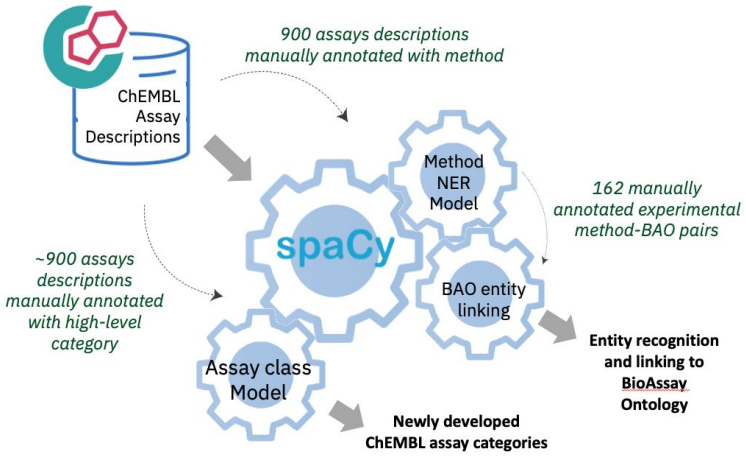
Improving Assay Annotation to Support AI-readiness



Assay description

Agonist activity at GAL4 DNA binding domain fused glucocorticoid receptor ligand binding domain (unknown origin) at 0.25 to 25 uM by: FRET-based assay **Experimental Method**

Named entity recognition	Ontology linking to BioAssay Ontology		Benefit
	BAO Preferred Term	BAO ID	
FRET-based assay	fluorescence resonance energy transfer	BAO_0000001	Specific annotation of assay features allows better comparison and grouping of assays for modelling. Current work focuses on experimental method, with future plans to expand other features e.g. setup, substrates, probes

New assay category	Ontology linking to BioAssay Ontology		Benefit
	BAO Preferred Term	BAO ID	
Protein activity	functional target-based	BAO_0013016	Highest level of assay classification hierarchy, designed to replace the current assay types

Refer to BAO hierarchy

- physical detection method
  - biosensor method
  - fluorescence method
    - flow cytometry
    - fluorescence intensity
    - fluorescence polarization
    - **fluorescence resonance energy transfer**
    - thermal shift

Smit I, Adasme MF, Manners E, Corbett S, Bosc N, Do H-M-A et al. **Enhancing Bioassay Annotations in ChEMBL with Artificial Intelligence.** *ChemRxiv.* **2025** doi:10.26434/chemrxiv-2025-h4hrr

EMBL-EBI

Curation of high-quality ML-ready data requires resource

What people imagine…

Curation of high-quality ML-ready data requires resource

What actually happens…

- Work with depositor to harmonise data format
- Request missing fields
- Extend ChEMBL data model
- Discussions on new data types
- Group data appropriately into assays and documents
- Fix typos, unusual characters

- Scientific validity: compare results to provided abstract (conversion errors)
- Encourage additional information to help with interpretability
- Check assignation of targets, and map to dictionaries
- Map other biological entities or variants present in assay to dictionaries
- Check for duplicates (and fix)
- Check assignation of compound names
- …

> 1.5 FTE / year over 5 years

Bioactivities and the patent literature

ChEMBL

Bioactivities from literature and deposited

Drug/candidate information

SureChEMBL

*Structures, **targets and diseases** from patents*

Bioactivities and the patent literature

ChEMBL

Bioactivities from literature and deposited

Drug/candidate information

SureChEMBL

*Structures, **targets and diseases** from patents*

Number of bioactivities extracted per (publication) year in ChEMBL versus extracted by LeadMine from US patents

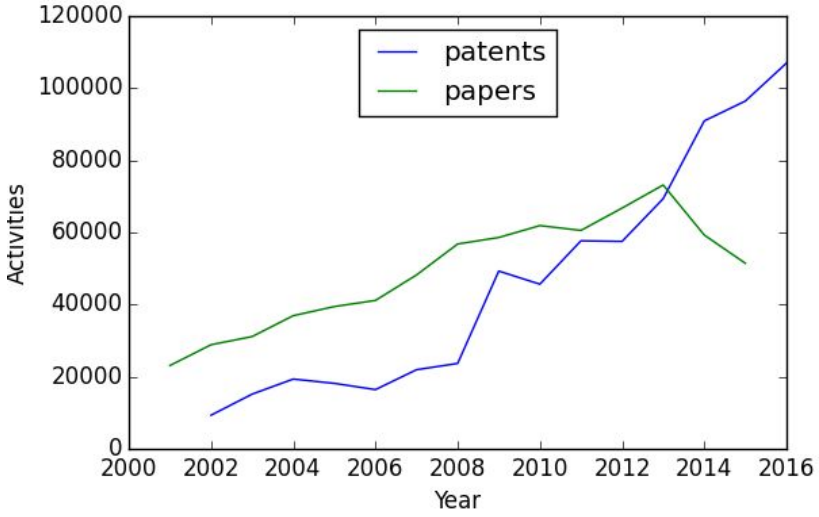


Image: https://nextmovesoftware.com/blog/2017/04/24/are-more-bioactivities-available-from-patents-than-from-the-academic-literature/

Bioactivities and the patent literature

ChEMBL

Bioactivities from literature and deposited

Drug/candidate information

SureChEMBL

Structures, ***targets, diseases and bioactivities*** *from patents*

Number of bioactivities extracted per (publication) year in ChEMBL versus extracted by LeadMine from US patents

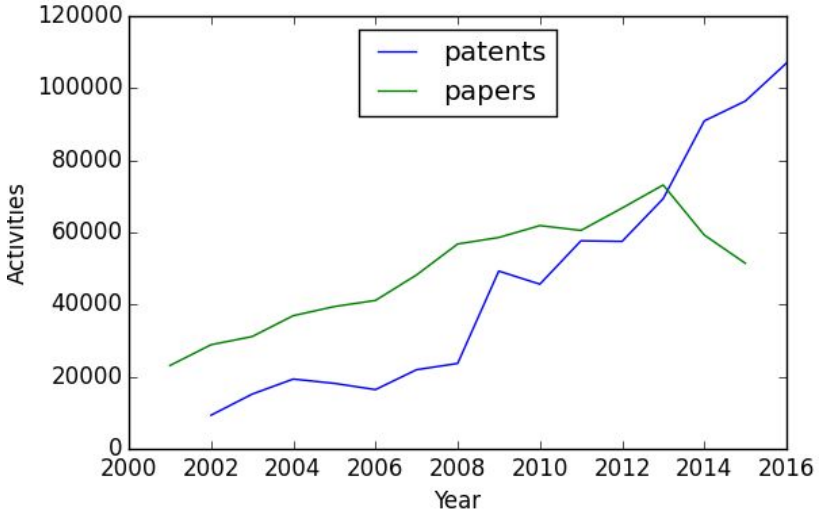


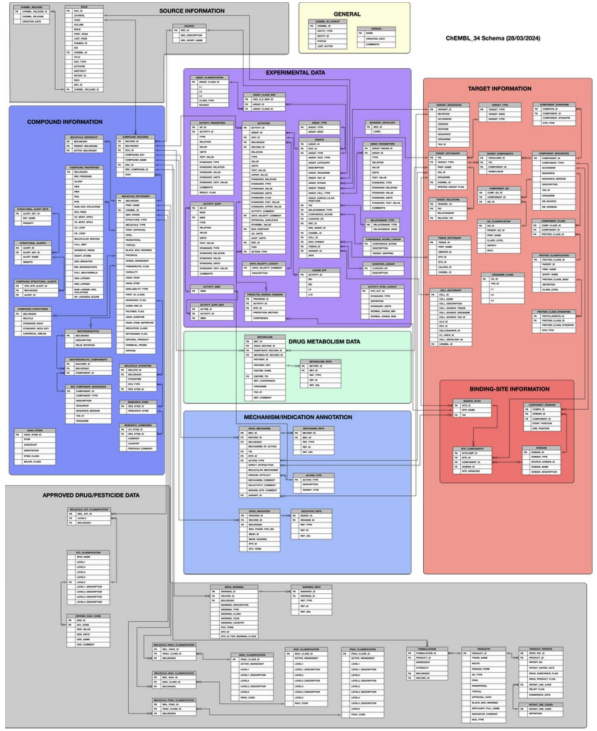
Image: https://nextmovesoftware.com/blog/2017/04/24/are-more-bioactivities-available-from-patents-than-from-the-academic-literature/

Data checks and curation are essential for high quality data to support machine learning

- Large-scale, open-access databases are a valuable source of drug discovery data, but some errors and noise are expected. ChEMBL provides references for all data points for data provenance.

- ChEMBL annotates bioactivity data with key features. Knowledge of the data structure and careful use of annotations is key.

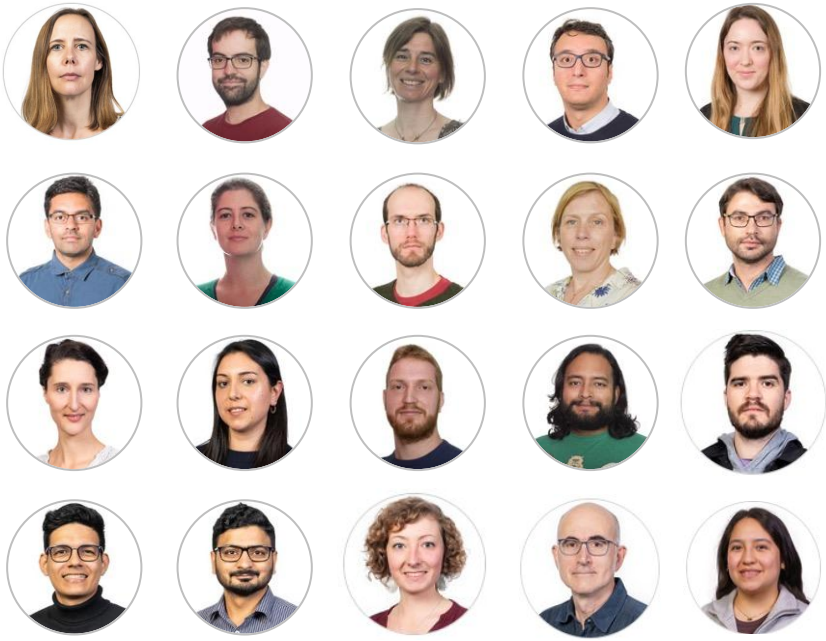
- Carefully-controlled data extraction protocols coupled with data clean-up can provide a more robust and explainable dataset, with clearly understood limitations.



ChEMBL Schema

EMBL-EBI

Acknowledgements



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Melanie Schneider

wellcome trust Strategic Award
Open Targets
imi innovative medicines initiative
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